IC637 Program Analysis

Lecture 3: Interval Domain

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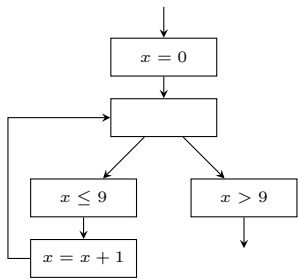
2025 Fall

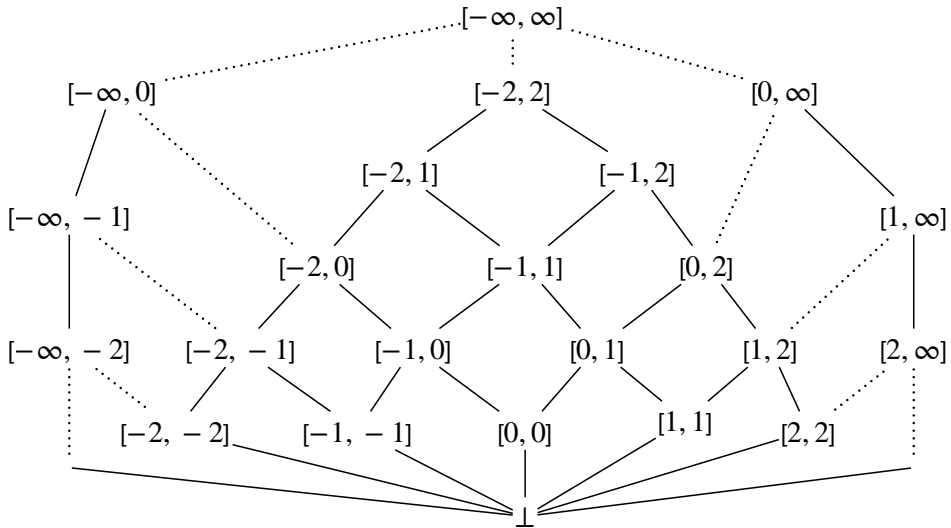
Example

- Can (extended) sign domain prove the assertion?
- How can we prove the assertion?

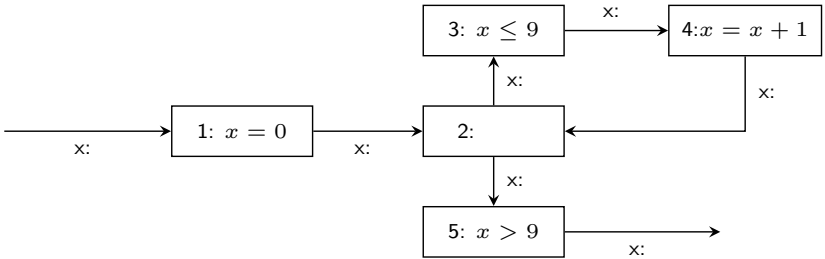
```
int main(){
    int x = 0;
    while(x <= 9){
        x = x + 1;
    }
    assert(x == 10);
    return 0;
}
```



Interval Domain

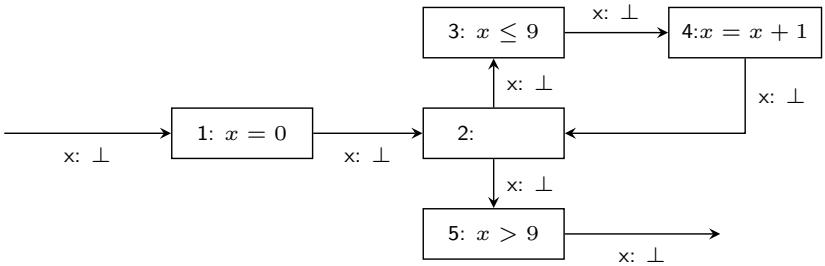


Fixed Point Computation



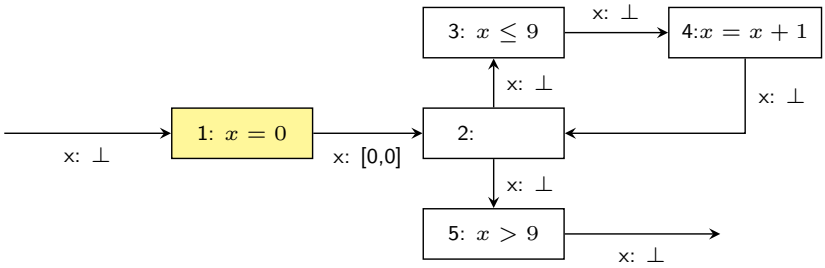
Worklist = $\{1, 2, 3, 4, 5\}$

Fixed Point Computation



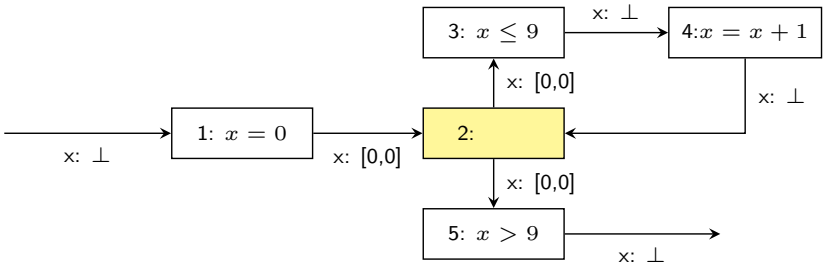
Worklist = {1, 2, 3, 4, 5}

Fixed Point Computation



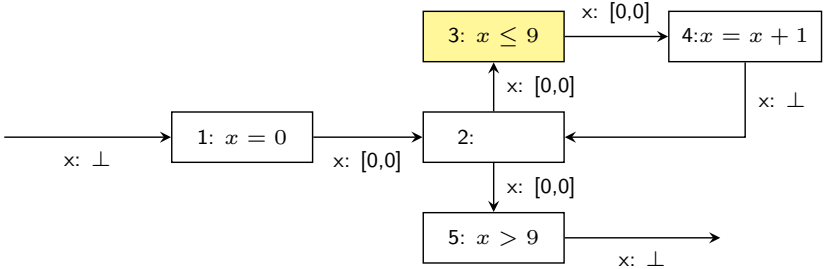
Worklist = $\{\not{1}, 2, 3, 4, 5\}$

Fixed Point Computation



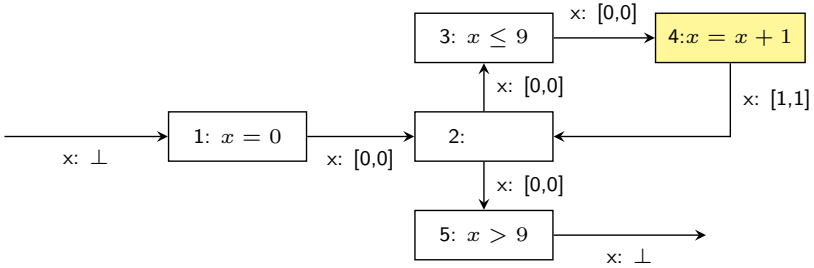
Worklist = $\{\cancel{1}, \cancel{2}, 3, 4, 5\}$

Fixed Point Computation



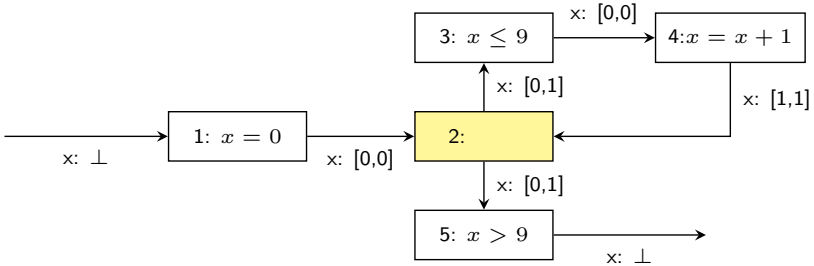
Worklist = {1, 2, 3, 4, 5}

Fixed Point Computation



Worklist $= \{\not{1}, 2, \not{3}, \not{4}, 5\}$

Fixed Point Computation

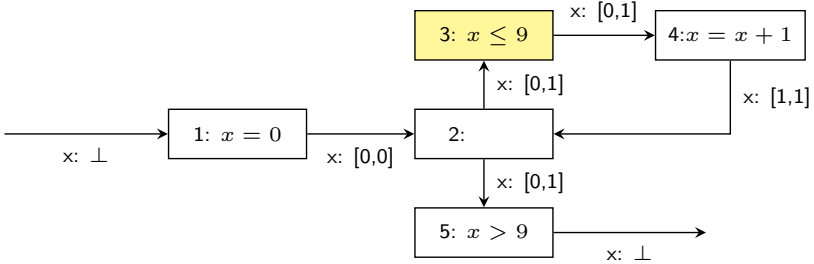


Input state : $[0,0] \sqcup [1,1] = [0,1]$

(1st iteration of loop)

Worklist = $\{1, 2, 3, 4, 5\}$

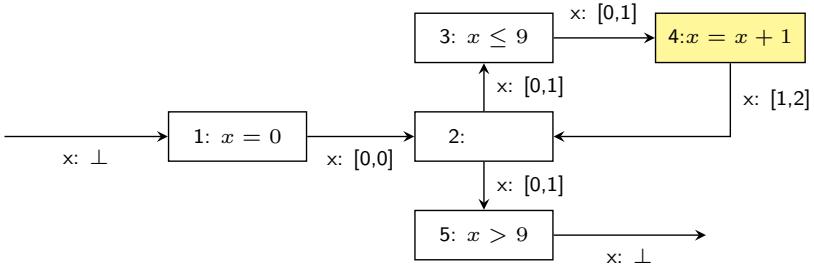
Fixed Point Computation



$$[0,1] \sqcap [-\infty, 9] = [0,1]$$

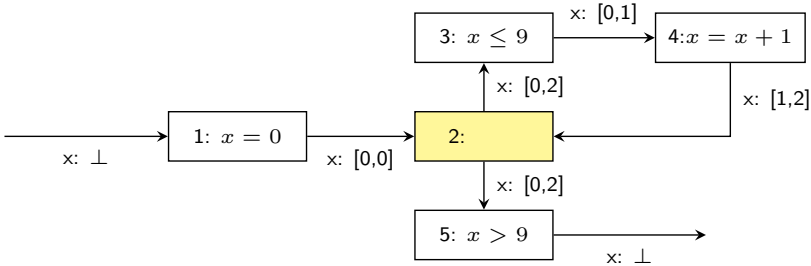
Worklist $= \{1, 2, 3, 4, 5\}$

Fixed Point Computation



Worklist $= \{1, 2, 3, 4, 5\}$

Fixed Point Computation

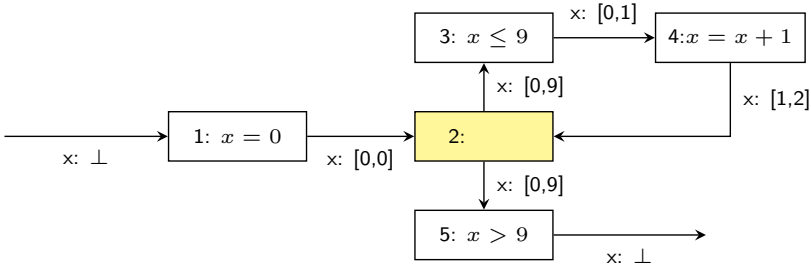


Input state : $[0,0] \sqcup [1,2] = [0,2]$

(2nd iteration of loop)

Worklist = {1, 2, 3, 4, 5}

Fixed Point Computation

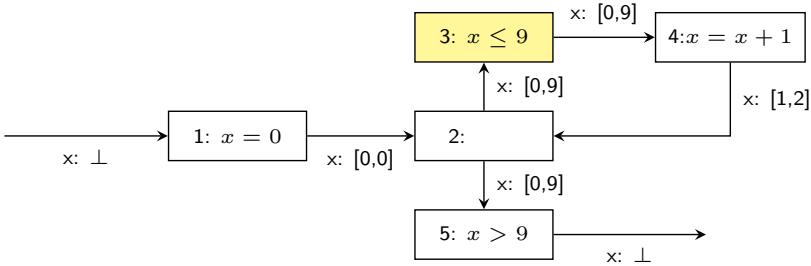


Input state : $[0, 0] \sqcup [1, 9] = [0, 9]$

(9th iteration of loop)

Worklist = $\{1, 2, 3, 4, 5\}$

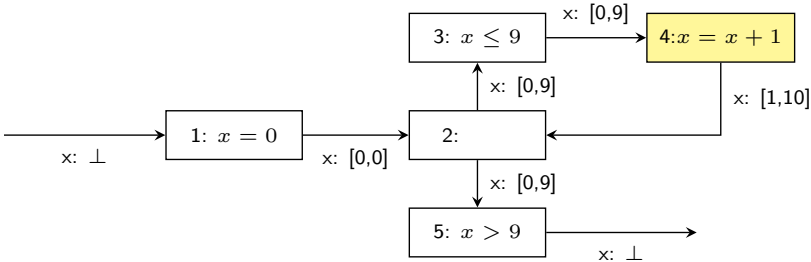
Fixed Point Computation



$$[0, 9] \sqcap [-\infty, 9] = [0, 9]$$

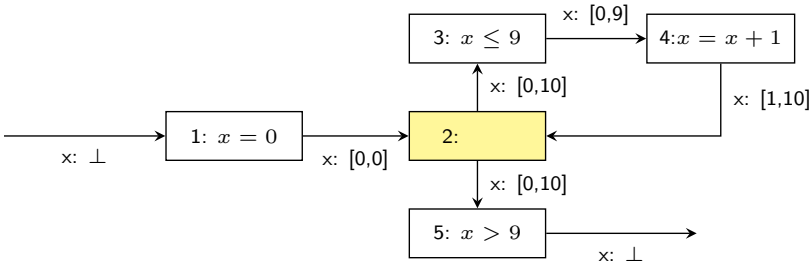
Worklist $= \{1, 2, 3, 4, 5\}$

Fixed Point Computation



Worklist $= \{1, 2, 3, 4, 5\}$

Fixed Point Computation

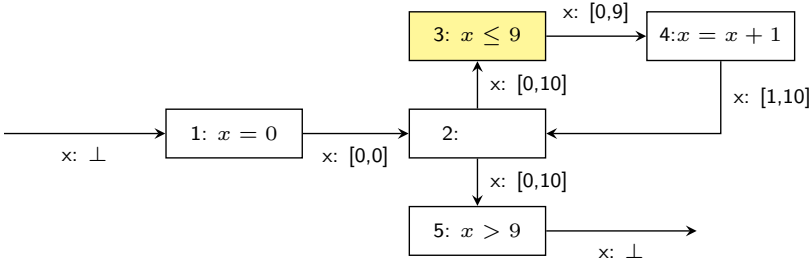


Input state : $[0, 0] \sqcup [1, 10] = [0, 10]$

(10th iteration of loop)

Worklist = $\{1, 2, 3, 4, 5\}$

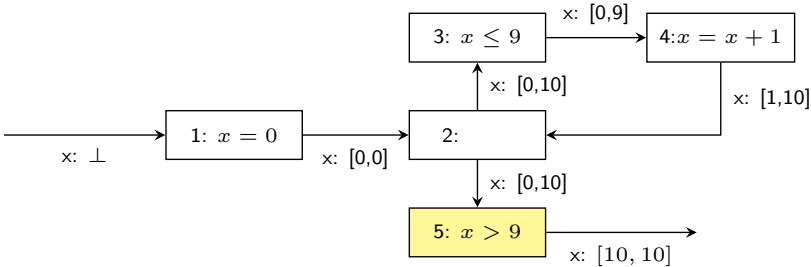
Fixed Point Computation



$$[0, 10] \sqcap [-\infty, 9] = [0, 9]$$

Worklist $= \{1, 2, 3, 4, 5\}$

Fixed Point Computation



$$[0, 10] \sqcap [-\infty, 10] = [10, 10]$$

Worklist $= \{1, 2, 3, 4, 5\}$

Interval Domain

- The set of intervals:

$$\hat{\mathbb{Z}} = \{\bot\} \cup \{[l, u] \mid l, u \in \mathbb{Z} \cup \{-\infty, \infty\}, l \leq u\}$$

- Partial order:

$$\bot \sqsubseteq \hat{z} \quad \text{(for any } \hat{z} \in \hat{\mathbb{Z}}\text{)}$$
$$[l_1, u_1] \sqsubseteq [l_2, u_2] \iff l_2 \leq l_1 \text{ and } u_1 \leq u_2$$

Interval Domain

- Addition / Subtraction / Multiplication:

$$[l_1, u_1] \hat{+} [l_2, u_2] =$$
$$[l_1, u_1] \hat{-} [l_2, u_2] =$$
$$[l_1, u_1] \hat{\times} [l_2, u_2] =$$

- Equality ($=$) produces $\top$ except for the cases:

$$[l_1, u_1] \hat{=} [l_2, u_2] = \hat{true} \quad (\text{if} \qquad)$$
$$[l_1, u_1] \hat{=} [l_2, u_2] = \hat{false} \quad (\text{if} \qquad)$$

- "Less than" ($<$) produces $\top$ except for the cases:

$$[l_1, u_1] \hat{<} [l_2, u_2] = \hat{true} \quad (\text{if} \qquad)$$
$$[l_1, u_1] \hat{<} [l_2, u_2] = \hat{false} \quad (\text{if} \qquad)$$

Interval Domain

- Join:

$$\bot \sqcup \hat{z} = \hat{z} \sqcup \bot = \hat{z} \quad \text{(for any } \hat{z} \in \hat{\mathbb{Z}}\text{)}$$
$$[l_1, u_1] \sqcup [l_2, u_2] = [\min(l_1, l_2), \max(u_1, u_2)]$$

- Meet:

$$\bot \sqcap \hat{z} = \hat{z} \sqcap \bot = \bot \quad \text{(for any } \hat{z} \in \hat{\mathbb{Z}}\text{)}$$
$$[l_1, u_1] \sqcap [l_2, u_2] = \begin{cases} [\max(l_1, l_2), \min(u_1, u_2)] & \text{if } \max(l_1, l_2) \leq \min(u_1, u_2) \\ \bot & \text{otherwise} \end{cases}$$

Abstract Memory

$$\hat{\mathbb{M}} = \mathsf{Var} \to \hat{\mathbb{Z}}$$

$$m_1 \sqsubseteq m_2 \iff \forall x \in \mathsf{Var}, m_1(x) \sqsubseteq m_2(x)$$

$$m_1 \sqcup m_2 = \lambda x. m_1(x) \sqcup m_2(x)$$

$$m_1 \sqcap m_2 = \lambda x. m_1(x) \sqcap m_2(x)$$

$$m_1 \nabla m_2 = \lambda x. m_1(x) \nabla m_2(x)$$

$$m_1 \triangle m_2 = \lambda x. m_1(x) \triangle m_2(x)$$

Question

- Give an example that the fixed point computation does not terminate.

Widening and Narrowing

- Widening:

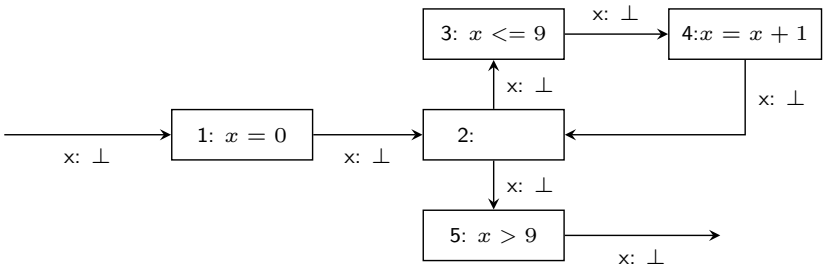
$$\bot \nabla \hat{z} = \hat{z} \nabla \bot = \hat{z} \quad \text{(for any } \hat{z} \in \hat{\mathbb{Z}})$$

$$[l_1, u_1] \nabla [l_2, u_2] = [l', u'] \text{ where}$$

$$l' = \begin{cases} l_1 & \text{if } l_2 \geq l_1 \\ -\infty & \text{otherwise} \end{cases}$$

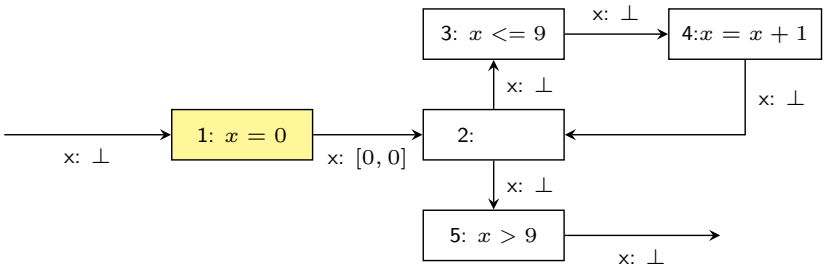
$$u' = \begin{cases} u_1 & \text{if } u_2 \leq u_1 \\ +\infty & \text{otherwise} \end{cases}$$

Fixed Point Computation with Widening



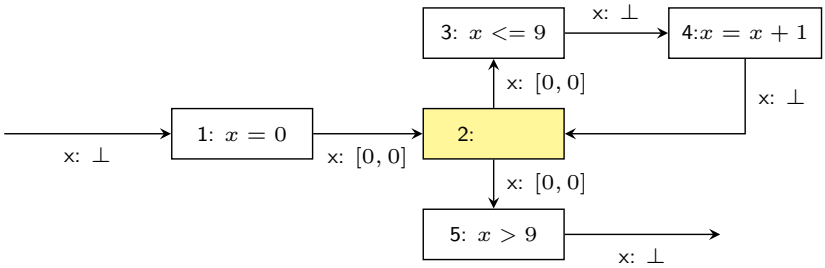
Worklist = {1, 2, 3, 4, 5}

Fixed Point Computation with Widening



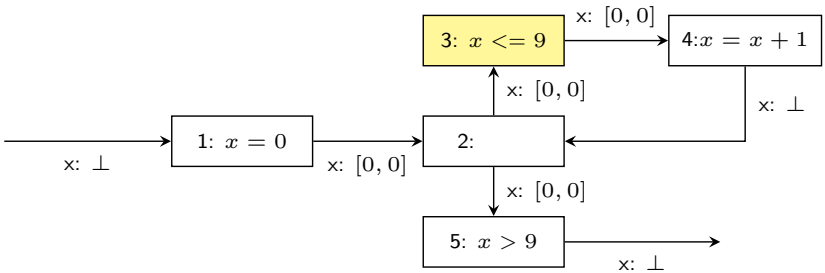
Worklist = $\{\not{1}, 2, 3, 4, 5\}$

Fixed Point Computation with Widening



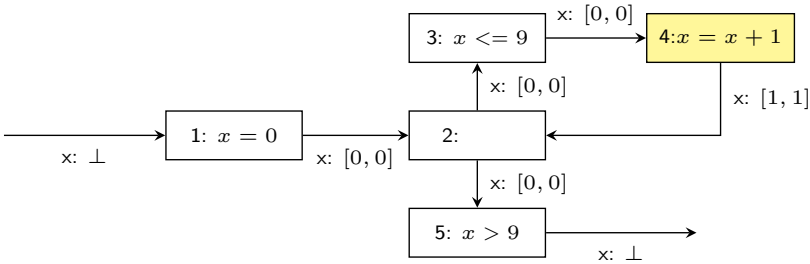
Worklist = $\{1, 2, 3, 4, 5\}$

Fixed Point Computation with Widening



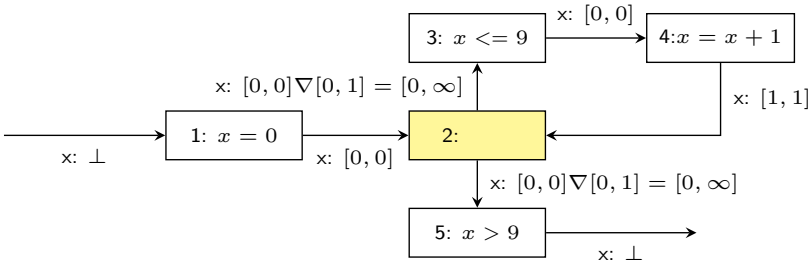
Worklist = $\{\cancel{1}, \cancel{2}, \cancel{3}, 4, 5\}$

Fixed Point Computation with Widening



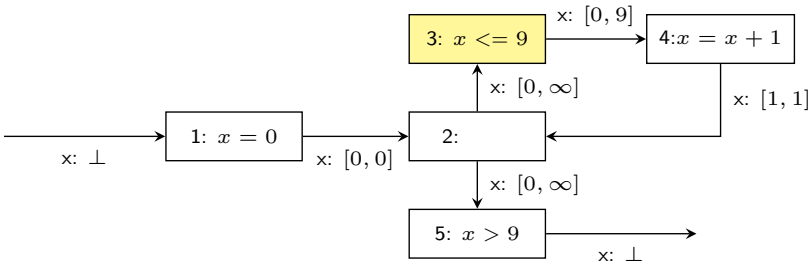
Worklist $= \{\not{1}, 2, \not{3}, \not{4}, 5\}$

Fixed Point Computation with Widening



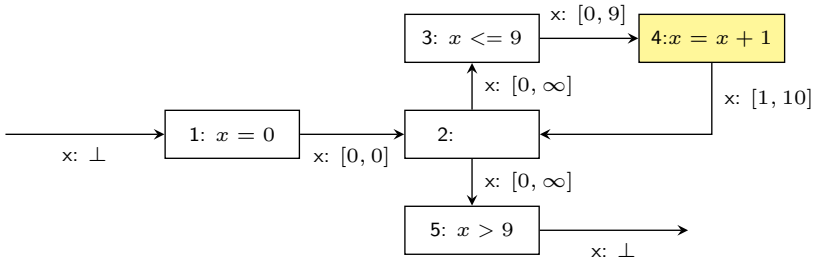
Worklist = $\{\cancel{1}, \cancel{2}, 3, \cancel{4}, 5\}$

Fixed Point Computation with Widening



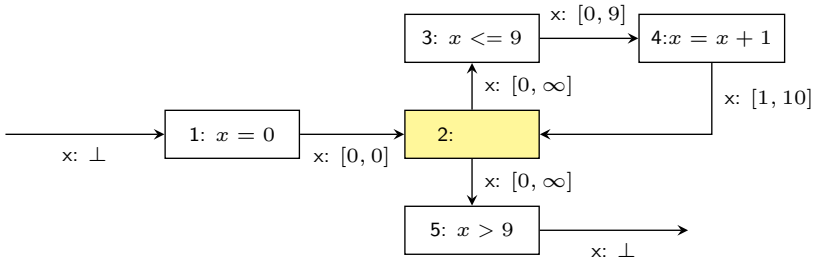
Worklist = $\{1, 2, 3, 4, 5\}$

Fixed Point Computation with Widening



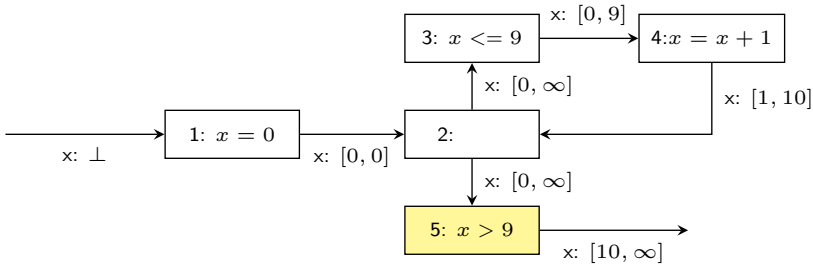
Worklist = $\{1, 2, 3, 4, 5\}$

Fixed Point Computation with Widening



Worklist = $\{1, 2, 3, 4, 5\}$

Fixed Point Computation with Widening



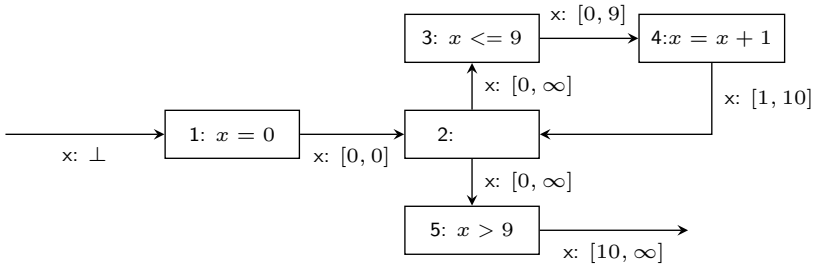
Worklist = $\{1, 2, 3, 4, 5\}$

Widening and Narrowing

- Narrowing:

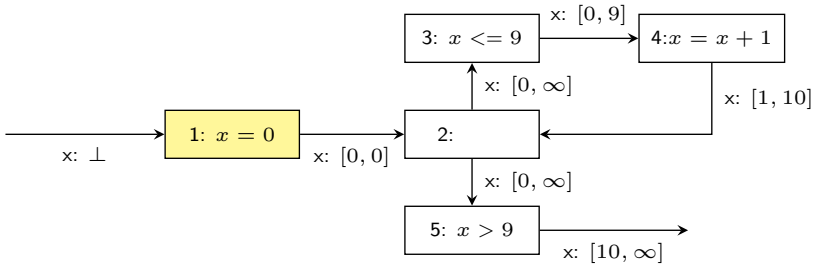
$$\bot \triangle \hat{z} = \bot \quad \text{(for any } \hat{z} \in \hat{\mathbb{Z}})$$
$$\hat{z} \triangle \bot = \bot \quad \text{(for any } \hat{z} \in \hat{\mathbb{Z}})$$
$$[l_1, u_1] \triangle [l_2, u_2] = [l', u'] \text{ where}$$
$$l' = \begin{cases} l_2 & \text{if } l_1 = -\infty \\ l_1 & \text{otherwise} \end{cases}$$
$$u' = \begin{cases} u_2 & \text{if } u_1 = +\infty \\ u_1 & \text{otherwise} \end{cases}$$

Fixed Point Computation with Narrowing



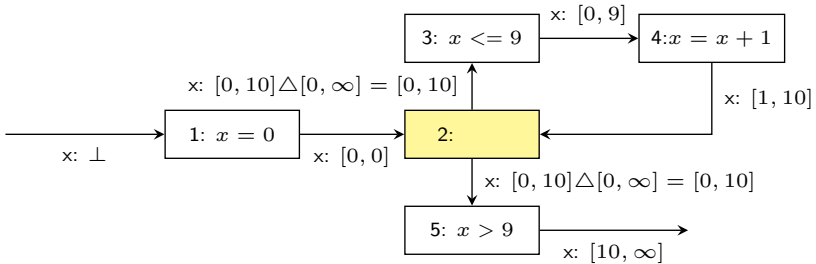
Worklist = {1, 2, 3, 4, 5}

Fixed Point Computation with Narrowing



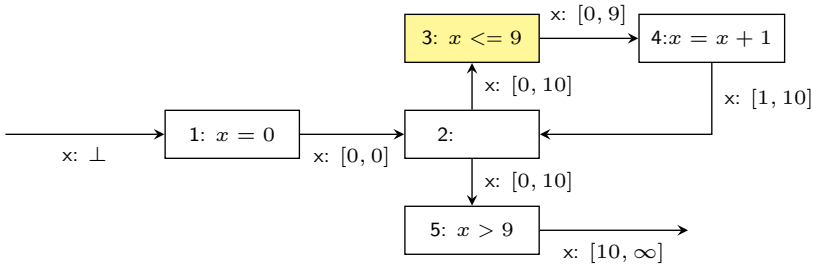
Worklist = $\{\cancel{1}, 2, 3, 4, 5\}$

Fixed Point Computation with Narrowing



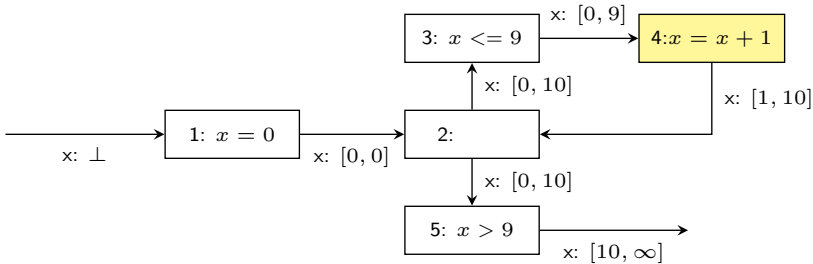
Worklist = $\{1, 2, 3, 4, 5\}$

Fixed Point Computation with Narrowing



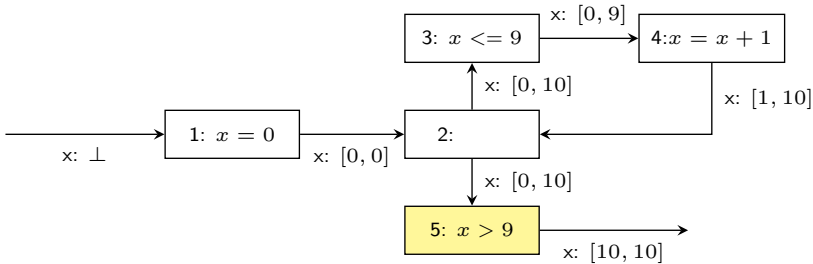
Worklist = {1, 2, 3, 4, 5}

Fixed Point Computation with Narrowing



Worklist = $\{1, 2, 3, 4, 5\}$

Fixed Point Computation with Narrowing



Worklist $= \{1, 2, 3, 4, 5\}$

Worklist Algorithm

Fixpoint comp. with Widening

$W := \textbf{Node}$
$T := \lambda n.\bot_{\hat{\mathbb{M}}}$
while $W \neq \emptyset$
  $n :=$ choose(W)
  $W := W \setminus \{n\}$
  $in :=$ inputof(n, T)
  $out :=$ analyze(n, in)
  if $out \not\sqsubseteq T(n)$ then
    if widening is needed
      $T(n) := T(n) \nabla out$
    else
      $T(n) := T(n) \sqcup out$
    $W := W \cup$ succ(n)

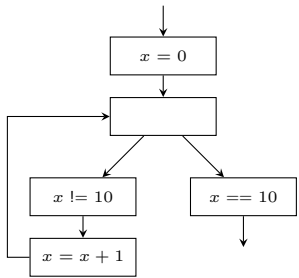
Fixpoint comp. with narrowing

$W := \textbf{Node}$
while $W \neq \emptyset$
  $n :=$ choose(W)
  $W := W \setminus \{n\}$
  $in :=$ inputof(n, T)
  $out :=$ analyze(n, in)
  if $T(n) \not\sqsubseteq out$ then
    $T(n) := T(n) \triangle out$
    $W := W \cup$ succ(n)

Example

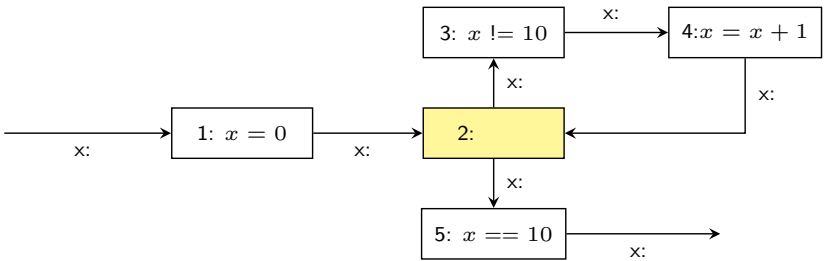
- Describe the result of the interval analysis:
  1. without widening
  2. with widening/narrowing

```
int main(){
    int x = 0;
    while (x != 10) {
        x = x + 1;
    }
    assert(x == 10);
    return 0;
}
```



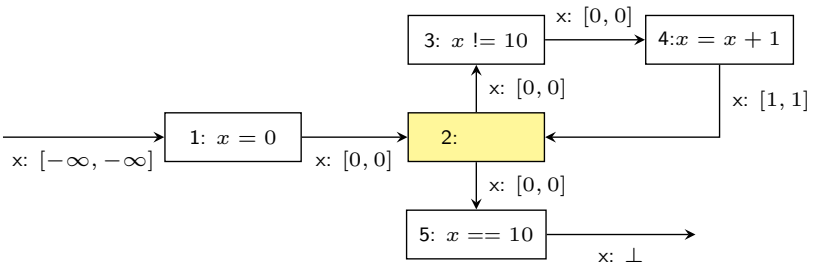
Widening with Thresholds

- Assume a set of thresholds $\theta = \{5, 10\}$ is given before the analysis



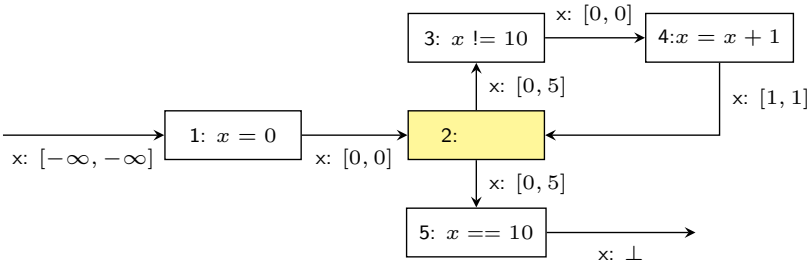
Widening with Thresholds

- Compute output by joining inputs : $[0,0] \sqcup [1,1] = [0,1]$



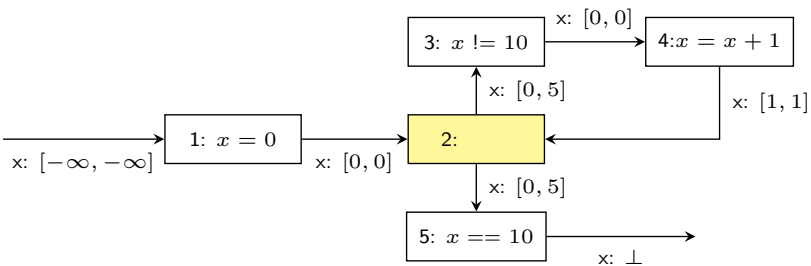
Widening with Thresholds

- Given $\theta = \{5, 10\}$, when applying widening: $[0, 0] \nabla [0, 1] = [0, 5]$

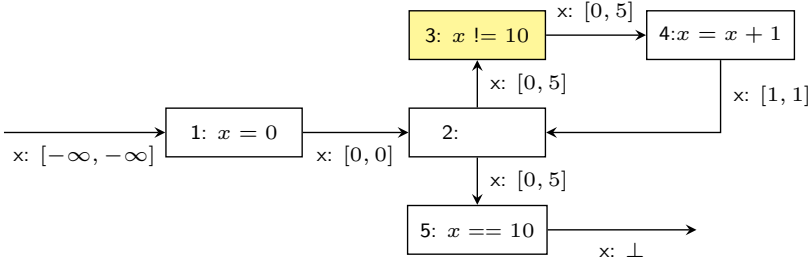


Widening with Thresholds

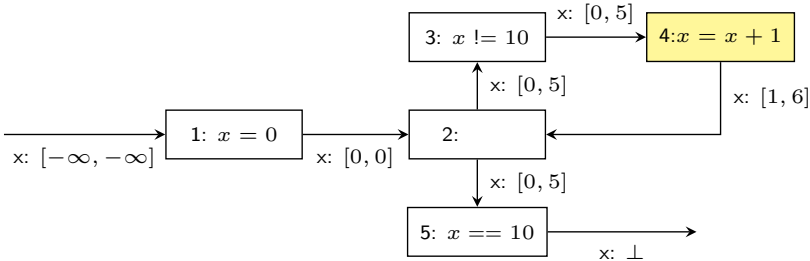
- Check fixed point is reached : $[0, 5] \not\sqsubseteq [0, 0]$



Widening with Thresholds

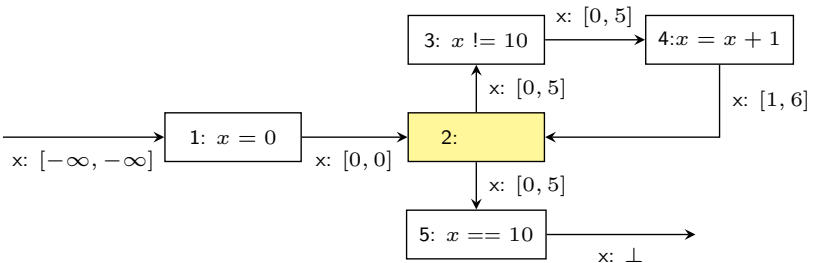


Widening with Thresholds



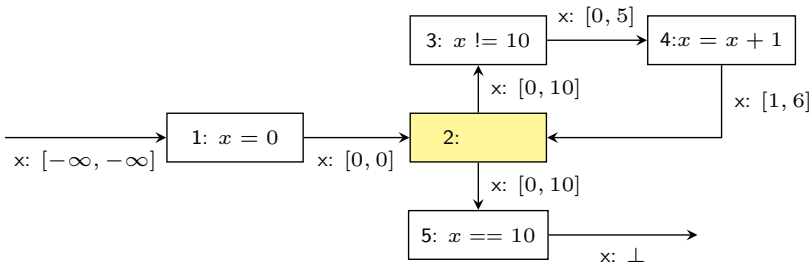
Widening with Thresholds

- Compute output by joining inputs: $[1, 6] \sqcup [0, 0] = [0, 6]$



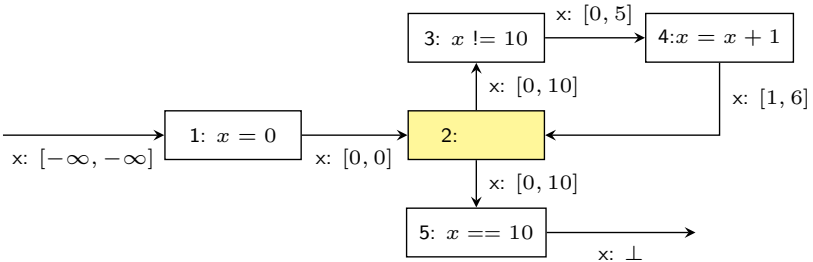
Widening with Thresholds

- Given $theta = \{5, 10\}$, use 10 as threshold when applying widening: $[0, 5] \nabla [0, 6] = [0, 10]$

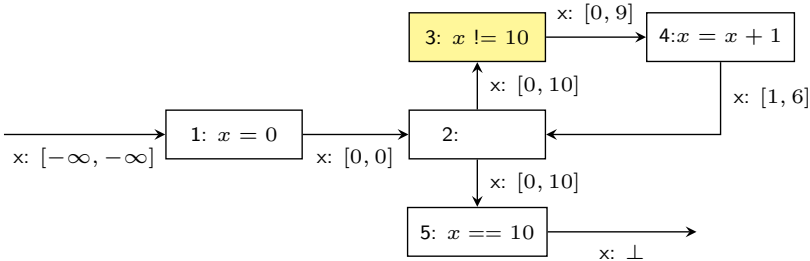


Widening with Thresholds

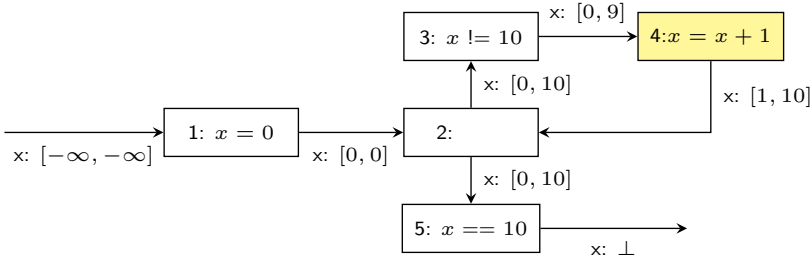
- Check if fixed point is reached: $[0, 10] \not\sqsubseteq [0, 5]$



Widening with Thresholds



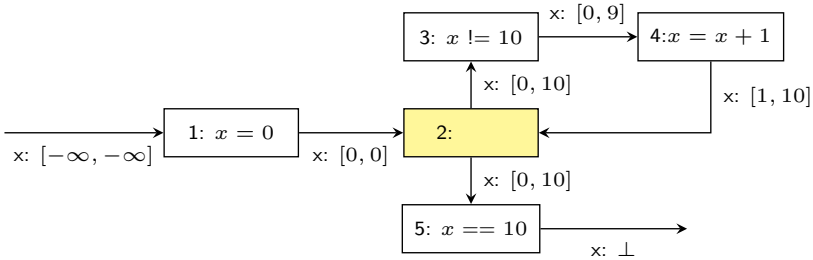
Widening with Thresholds



Widening with Thresholds

- Compute output by joining inputs:

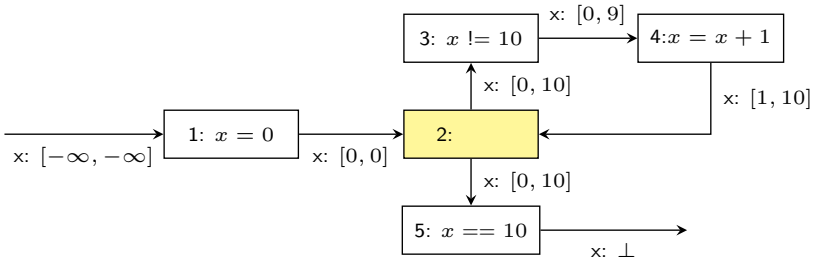
$$[0, 0] \sqcup [1, 10] = [0, 10]$$



Widening with Thresholds

- Apply widening:

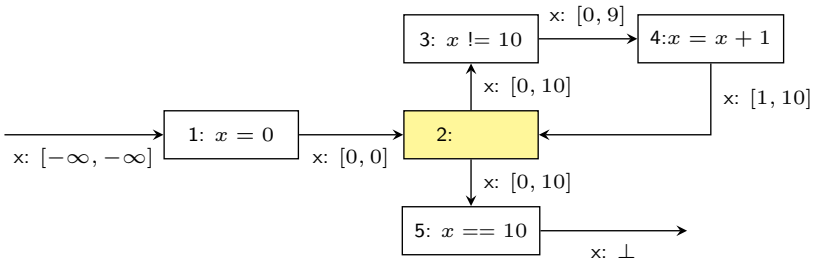
$$[0, 10] \nabla [0, 10] = [0, 10]$$



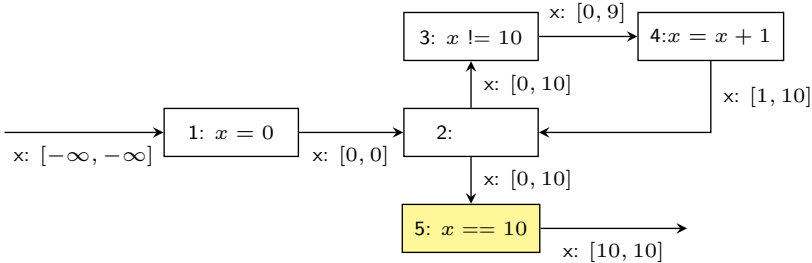
Widening with Thresholds

- Check if fixed point is reached:

$$[0, 10] \sqsubseteq [0, 10]$$



Widening with Thresholds



Widening with Thresholds

- A threshold set $\theta \subseteq \mathbb{Z}$ is given.

$$\bot \nabla_\theta \hat{z} = \hat{z}$$

$$\hat{z} \triangle_\theta \bot = \hat{z}$$

$$[l_1, u_1] \nabla_\theta [l_2, u_2] = [l_3, u_3] \text{ where}$$

$$l_3 = \begin{cases} glb(\theta, l_2) & \text{if } l_1 > l_2 \\ l_1 & \text{otherwise} \end{cases}$$

$$u_3 = \begin{cases} lub(\theta, u_2) & \text{if } u_1 < u_2 \\ u_1 & \text{otherwise} \end{cases}$$

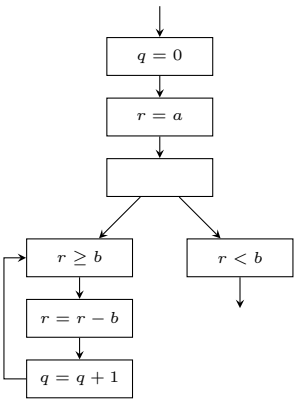
$$glb(\theta, n) = \max\{t \in \theta \mid t \leq n\}$$

$$lub(\theta, n) = \min\{t \in \theta \mid t \geq n\}$$

Excercise

- Describe the result of the interval analysis with widening and narrowing

```
//a >= 0, b >= 0
q = 0;
r = a;
while(r >= b){
    r = r - b;
    q = q + 1;
}
assert(q >= 0);
assert(r >= 0);
```



Summary

- **Interval Domain**: Abstract domain $\hat{\mathbb{Z}} = \{\bot\} \cup \{[l, u] \mid l, u \in \mathbb{Z} \cup \{-\infty, \infty\}, l \leq u\}$
  - Partial order: $\bot \sqsubseteq x$, $[l_1, u_1] \sqsubseteq [l_2, u_2]$ iff $l_2 \leq l_1$ and $u_1 \leq u_2$
  - Join: $[l_1, u_1] \sqcup [l_2, u_2] = [\min(l_1, l_2), \max(u_1, u_2)]$
  - Meet: $[l_1, u_1] \sqcap [l_2, u_2] = [\max(l_1, l_2), \min(u_1, u_2)]$ if overlap, $\bot$ otherwise
- **Worklist Algorithm**: Iterative fixed-point computation
  - Widening phase: Apply widening at loop headers until convergence
  - Narrowing phase: Apply narrowing to refine results
- **Widening & Narrowing**: Essential for termination in infinite domains
  - Widening (∇): Extrapolates to infinity to ensure convergence
  - Narrowing ($\triangle$): Refines over-approximations from widening
  - Widening with thresholds: Uses predefined values to improve precision
- **Key Insight**: Balance between precision and scalability (e.g., termination)